Prime geodesic theorem for arithmetic compact surfaces

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Outline

What is PGT

Proof of PGT

Prime number theorem

In the 18th century, people first discovered that the asymptotic distribution of prime numbers follows a simple law:

$$\pi(x) = \#\{p \text{ prime} : p \leq x\} \sim \frac{x}{\log x}.$$

This is the famous prime number theorem (PNT).

Proof of PNT

About a hundred years later, Hadamard and Poussin independently proved this.
The proof contains three steps.

- Prove that PNT is equivalent to this statement.

$$\psi(x) = \sum_{n \leq x} \Lambda(n) = \sum_{p^e \leq x, p \text{ prime}} \log p \sim x.$$

- Prove that Riemann zeta function $\zeta(s)$ has no zeros on the line $\Re(s) = 1$.
- Using Mellin transformation, Write $\psi(x)$ as an integral, which is x plus something related to the Riemann zeta function.

Error terms in PNT

In fact, we can write the error terms in PNT explicitly.

$$\psi(x) = \int_{c-i\infty}^{c+i\infty} \frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s} ds = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - \frac{1}{x^2}).$$

The sum takes ρ over all the nontrivial zeros of $\zeta(s)$. The famous Riemann hypothesis (RH) says that all of $\Re(\rho) = \frac{1}{2}$.
RH is a notoriously hard open problem in mathematics. In fact one can even not prove that $\Re(\rho) < 1 - \epsilon$ for any $\epsilon > 0$ now.

Prime geodesic theorem

Prime geodesic theorem (PGT) is just an analogy to PNT. The first PGT is proved for $PSL_2(\mathbb{Z})\backslash\mathbb{H}$. Here $\mathbb{H}$ is the upper half plane. One can generalize to other arithmetic compact hyperbolic surfaces, e.g. $\mathbb{H}$ modulo congruence subgroups.

$\Gamma = PSL_2(\mathbb{Z})\backslash\mathbb{H}$

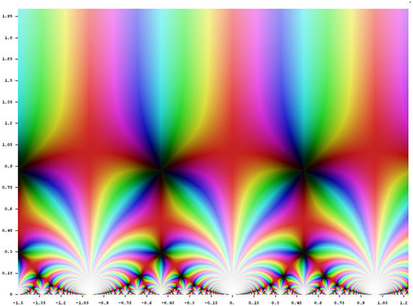

Figure 1: Image of j function, which is a typical function on Γ

PNT vs PGT

- Prime numbers.
- Norm of a prime number is just its value.
- $\pi(x) = \int_0^x \frac{dt}{\log t} + E(x), \psi(x) = x + \tilde{E}(x)$.
- Riemann zeta function $\zeta(s)$.
- RH.

- Prime geodesics on the hyperbolic surface Γ.
- Norm of a prime geodesic is the logarithm of its length.
- $\pi_\Gamma(x) = \int_0^x \frac{dt}{\log t} + E(x), \Psi_\Gamma(x) = x + \tilde{E}(x)$.
- Selberg zeta function $Z_\Gamma(s)$.
- RH.

The philosophy

There are some crucial differences in PGT compared with PNT.

- GL_2 is non-abelian, thus Selberg trace formula is needed.
- $Z_\Gamma(s)$ is of order 2, unlike $\zeta(s)$ which is of order 1. So $Z_\Gamma(s)$ contains more zeros. Now RH only implies that $E(x) \ll x^{3/4+\epsilon}$, not $E(x) \ll x^{1/2+\epsilon}$.
- As you may guess from the previous difference, now RH is much easier to handle.

Progress

- (Iwaniec, 1984) $E(x) \ll x^{35/48+\epsilon}$.
- (Luo-Sarnak, 1995) $E(x) \ll x^{7/10+\epsilon}$.
- (Soundararajan-Young, 2013) $E(x) \ll x^{25/36+\epsilon}$.

PGT for quaternions

Koyama first extended the Luo-Sarnak bound to the surface $\Gamma_{\mathcal{D}}$ from a quaternion algebra $\mathcal{D}$ (Koyama, 1998).
Koyama's method exploits the Jacquet-Langlands correspondence (JL), but working on the spectral side.

Quaternions

Quaternions for real numbers are just generalizations of complex numbers, where you use 3 imaginary unit i, j, k satisfying associative law with the following relations.

$$i^2 = j^2 = k^2 = ijk = -1.$$

But now you have $ij = k = -ji$. For general fields the construction is similar.

Jacquet-Langlands correspondence

There are two types of degree 2 central simple algebra over a field, matrix algebra M_2 and quaternion algebra $\mathcal{D}$. Jacquet-Langlands correspondence says that their representations have some relations.

New work

We generalize this result to principal subgroups $\Gamma_{\mathcal{D}}(\mathfrak{N})$, using also JL but working on the arithmetic side (Tang-Wu-Yang-Y., 2026).

Proof Sketch

The proof contains 5 steps.

- Like the proof of PNT, write Ψ as an integral.

Integral expression of Ψ

$$I(t;\widetilde{f}_\infty) := \int_{\Gamma^1\mathbb{R}^\times\backslash GL_2(\mathbb{R})} \left(\sum_{\sigma\in[\gamma]_{\mathrm{st}}} \widetilde{f}_\infty(g^{-1}\sigma g) \right) dg = \frac{\widetilde{g}_\infty(\log x)}{\sqrt{x}} \cdot \psi_{\Gamma^1}(t)\,.$$

Here $x = \frac{|t|+\sqrt{t^2-4}}{2}$.

Proof Sketch

The proof contains 5 steps.

- Like the proof of PNT, write Ψ as an integral.
- Turn it into an adelic orbital integral.

Adelic integral

$$I(t;\widetilde{f}_\infty) = (\text{some constant}) \int_{\mathbf{Z}_\gamma(\mathbb{A})\backslash\mathcal{B}^\times(\mathbb{A})} \left(\widetilde{f}_\infty \otimes id_{\mathbf{K}_\Gamma}\right)(g^{-1}\gamma g)dg.$$

The constant is

$$\frac{1}{c_{\Gamma^1}} \frac{Vol(\widehat{\mathbb{Z}}^\times)}{[\widehat{\mathbb{Z}}^\times : \nu(\mathbf{K}_\Gamma)]} \frac{Vol\left(\mathbf{Z}_\gamma(\mathbb{Q})\mathbb{A}^\times\backslash\mathbf{Z}_\gamma(\mathbb{A})\right)}{Vol(\mathbf{K}_\Gamma)}.$$

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- Match the local orbital integrals of the matrix algebra and corresponding quaternions. (key part)

Key part

This part essentially is a concrete version of JL. The difference is that we find the right test functions and check they do give the same integral value.

$$\begin{cases} \widetilde{f_0} := 2f_0 - g_0 \leftrightarrow \varphi_0 & \\ \widetilde{f_{2n}} := \frac{2q}{q-1} f_n - \frac{q+1}{q-1} g_{2n} \leftrightarrow \varphi_{2n} & \forall\ n \in \mathbb{Z}_{\geq 1} \\ \widetilde{f_{2n-1}} := -\frac{2}{q-1} f_n + \frac{q+1}{q-1} g_{2n-1} \leftrightarrow \varphi_{2n-1} & \forall\ n \in \mathbb{Z}_{\geq 1} \end{cases}.$$

Key part

The meaning of f_n, g_n, φ_n is as follows.

$$f_n := \frac{1}{[\mathfrak{o}^\times : \det U^n_{\mathfrak{M}}]} \cdot \frac{1}{Vol(U^n_{\mathfrak{M}})} 1_{U^n_{\mathfrak{M}}},$$

$$g_n := \frac{1}{[\mathfrak{o}^\times : \det U^n_{\mathfrak{J}}]} \cdot \frac{1}{Vol(U^n_{\mathfrak{J}})} 1_{U^n_{\mathfrak{J}}},$$

$$\varphi_n := \frac{1}{[\mathfrak{o}^\times : \nu_{\mathcal{D}}\big(U^n_{\mathcal{D}}\big)]} \cdot \frac{1}{Vol(U^n_{\mathcal{D}})} 1_{U^n_{\mathcal{D}}}.$$

Proof Sketch

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- Like the proof of PNT, write Ψ as an integral.
- Turn it into an adelic orbital integral.
- Write adelic orbital integral as product of local orbital integrals.
- Match the local orbital integrals of the matrix algebra and corresponding quaternions. (key part)
- Combined Luo-Sarnak's bound, PGT is true in this case. $\square$

THANK YOU